CEPH PRODUCTION BENCHMARK AUDIT REPORT

RADOS Performance, Integrity, Health and Cleanup Validation

Document owner	Nishu Saini
Function	CISAI IT Support / Infrastructure
Cluster	cisai-pve
Execution host	prox24
Evidence file	benchmark_ceph.log
Validation date	05 September 2026
Classification	Internal - Production Audit Evidence
Document version	1.0

OVERALL AUDIT RESULT: PASS

Two executed Ceph benchmark runs completed with 17 passes, 0 warnings and 0 failures. Ceph remained HEALTH_OK and temporary test pools were removed.

This report summarizes the supplied system-generated benchmark log. The original log remains the authoritative detailed evidence.

1. Executive Summary

The supplied log contains one pre-flight-only run and two executed RADOS benchmark runs on prox24. Both executed runs completed write, sequential-read, random-read, object-integrity, post-test health and cleanup stages successfully. The cluster was quorate before testing, all OSDs were up and in, placement groups were active+clean, and Ceph remained HEALTH_OK after testing.

Control	Recorded evidence	Disposition
Pre-flight	Three-node cluster quorate; HEALTH_OK; all OSDs up/in; PGs active+clean	PASS
60-second run	16 concurrent I/O; 32 PG temporary pool; all benchmark phases completed	PASS
30-second run	4 concurrent I/O; 32 PG temporary pool; all benchmark phases completed	PASS
Object integrity	Write/read comparisons succeeded; paired SHA-256 values matched	PASS
Post-test health	HEALTH_OK after both executed tests	PASS
Cleanup	8,600 and 2,937 benchmark objects removed; dedicated pools removed	PASS

Audit conclusion

The supplied evidence demonstrates successful controlled RADOS benchmark execution and object-integrity verification without a recorded Ceph health failure. The results are point-in-time measurements under the exact test parameters recorded below.

2. Test Scope and Parameters

Run	Start time	Duration	Concurrent I/O	PGs	Pool
Pre-flight only	09:45:59 +05:30	No test I/O	N/A	N/A	ceph-benchmark-20260905-094559
Executed run A	09:46:27 +05:30	60 seconds	16	32	ceph-benchmark-20260905-094627
Executed run B	09:50:10 +05:30	30 seconds	4	32	ceph-benchmark-20260905-095010

The first run intentionally stopped after pre-flight checks and recorded PASS WITH WARNINGS because real test I/O was not requested. This is not treated as a benchmark failure.

3. Baseline Cluster and Ceph Health

Item	Recorded state	Result
Cluster	cisai-pve; 3 nodes; Quorate: Yes	PASS
Ceph health	HEALTH_OK	PASS
OSDs	12 total; all up and in	PASS
Placement groups before tests	33 active+clean	PASS
Raw capacity	218 TiB total and available in baseline output	RECORDED
Ceph RBD storage	Active in pvesm status	PASS
MON quorum	prox24, prox22 and prox23	PASS
MGR	prox23 active; prox24 and prox22 standby	PASS

4. Benchmark Results

4.1 Executed Run A - 60 seconds, 16 concurrent I/O

Workload	Bandwidth	Average IOPS	Average latency	Maximum latency
Write	571.424 MB/s	142	0.111720 s	0.657908 s
Sequential read	872.889 MB/s	218	0.0730442 s	0.687909 s
Random read	865.944 MB/s	216	0.0736673 s	0.860242 s

4.2 Executed Run B - 30 seconds, 4 concurrent I/O

Workload	Bandwidth	Average IOPS	Average latency	Maximum latency
Write	389.027 MB/s	97	0.0409503 s	0.432513 s
Sequential read	431.741 MB/s	107	0.036881 s	0.655107 s
Random read	462.498 MB/s	115	0.034427 s	0.401816 s

Interpretation note

These are recorded RADOS results for 4 MiB objects under the stated concurrency and duration. The report does not treat the values as a service-level objective, guarantee or long-term baseline because the supplied evidence contains point-in-time benchmark executions only.

5. Integrity, Post-Test Health and Cleanup

Control	Run A evidence	Run B evidence	Result
Object comparison	Succeeded	Succeeded	PASS
SHA-256	70417e... values matched	e7300a... values matched	PASS
Ceph health after test	HEALTH_OK	HEALTH_OK	PASS
PG state after test	65 active+clean	65 active+clean	PASS
OSD latency output	All 12 commit/apply values shown as 0 ms	All 12 commit/apply values shown as 0 ms	PASS
Objects removed	8,600	2,937	PASS
Temporary pool removed	ceph-benchmark-20260905-094627	ceph-benchmark-20260905-095010	PASS
Final test summary	17 PASS, 0 WARN, 0 FAIL	17 PASS, 0 WARN, 0 FAIL	PASS

6. Audit Observations

ID	Observation	Disposition
INFO-01	The pre-flight-only run recorded one warning because --execute was not used. No test I/O occurred in that run.	Informational
INFO-02	Node-local ZFS stores for prox22 and prox23 appeared disabled when the status was collected on prox24.	Recorded context; not evaluated as a Ceph benchmark defect
OBS-01	The evidence is a RADOS benchmark from prox24. The log does not provide a simultaneous application-level VM latency baseline.	Retain as scope limitation
OBS-02	Results differ between concurrency 16 and concurrency 4, as recorded. No pass/fail threshold was supplied for performance values.	Use as baseline evidence, not SLA validation

7. Audit Sign-off

Sign-off acknowledges review of the summarized results and stated scope limitations.

Role	Name	Signature	Date
Prepared by	Nishu Saini		
Technical reviewer			
Change owner / approver			

AUDIT DISPOSITION: PASS

Both executed Ceph production benchmark runs completed successfully, preserved HEALTH_OK, verified object integrity and removed their dedicated test pools.

Appendix A - Evidence Traceability

Claim	Evidence location in benchmark_ceph.log
Cluster quorum and baseline health	PRE-FLIGHT VALIDATION sections
Run A metrics	RADOS WRITE, SEQUENTIAL READ and RANDOM READ sections for pool ending 094627
Run B metrics	RADOS WRITE, SEQUENTIAL READ and RANDOM READ sections for pool ending 095010
Integrity verification	OBJECT INTEGRITY TEST sections
Post-test health	POST-TEST HEALTH sections
Pool/object cleanup	CLEANUP sections
Final disposition	FINAL SUMMARY sections

Appendix B - Evidence Retention

Retain this PDF together with the original benchmark_ceph.log and the individual report files referenced by the log. Preserve the original SHA-256 values, filenames and timestamps. Store future reruns as separate evidence sets.